

AI Driven Fiscal Planning through Budget Forecasting and Allocation using Time Series and Gradient Boosting Models

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Abstract - Efficient budget allocation is a challenge that government agencies often face. Most budget decisions are made in traditional processes based on past experiences, sometimes ignoring changing economic and social conditions. This paper presents an Artificial Intelligence-based framework that aims to improve budget allocation in the Indian public sector. It contains the information about the Union Budget press releases of the years 2016- 2024, the macroeconomic variables such as GDP, inflation, unemployment, and the level of literacy, and explains the proposed system that includes budget and econometric modelling. The proposed system combines the time series forecasting methods (ARIMA and Prophet) with machine learning regression methods (XGBoost and Random Forest). This combination takes into consideration the budget needs of each sector while spotting both over and underspend trends, and all future expenditure plans based on trends. To enhance the government financial agency prediction engine's forecasting accuracy, we included feature engineering elements consisting of lagged financial measurements, macroeconomic measurements, and election year impacts. The scenario simulation enables policymakers to adjust economic assumptions facilitating a range of budget possibilities. In terms of setting the aggregate budgetary expenditure, the model of optimal allocation employs priority ranking weight adjustment model which incorporates policy intentions and monetary limits. The experimental results indicate that the proposed model is successful in both predicting accuracy and lessening budgetary expenditure inefficiencies, while providing a specific set of recommendations through an engaged dashboard interface. The model serves as a comprehensive data analytics and modeling approach providing a means in which to improve the accountability, transparency, and efficiency of public sector financial management

Keywords — Budget, predictive analytics, time-series, socio-economic, government budgeting, policy-driven, machine learning, ARIMA, Prophet, XGBoost, Random Forest, government budgeting, scenario plan, fiscal plan, resource optimization, policy-driven allocation, economical forecasting.

I. INTRODUCTION

Budget allocation represents a critical aspect of public financial management, as it determines how national resources are allocated across public sectors like Healthcare, education, roads, public infrastructure and defence. The success of these allocations directly influences socio-economic development, the level of public services provided and the efficiency level of governance. In India, the most important fiscal document is the Union Budget which expresses the Government priorities in the upcoming financial year. The budgeting process uses previous spending patterns, policy priorities and politics to create the budget, but, does not ensure or guarantee equitable and efficient

distribution of funds. Standard budgeting methods stress marginal changes over the previous budget year and, sectors with increasing demand tend to be treated like slow-growth sectors and are thus underfunded, whereas slow-growth sectors receive funds more than their actual need. Conventional budgeting processes are slow to respond to the emergent challenges like economic recessions, epidemiological conditions or natural disasters. Also, the standard focusing method does not use socio-economic indicators as inputs—GDP growth, inflation trends, unemployment statistics, and literacy levels— when making funding decisions. This creates lost opportunities of data-based planning.

New trends in Artificial Intelligence and Machine Learning will relatively transform traditional budget allocation practices. Various time-series forecasting models, including ARIMA and Prophet will have the capability to model past sector trends and seasonality of underfunding. When used in connection with regression and ensemble algorithms (e.g., XGBoost, Random Forest), richer socio-economic data can be brought into the forecast, which would ultimately leads to better forecasts about budgets, detecting inefficiencies, and or modelling various policy or economic alternatives to develop more flexible decision-making processes.

Despite an uptick in using AI for current to forecasted financial metrics, there are practically very few studies that identify endogenous budget development trends combined with their influences of socio-economic indicators in one predictive model for public sector budget development in India. The current use of forecasting methods tend to be either user major economic forecasts or by very short-term monitored spending, leaving a appropriate knowledge transfer gap within any policy relevant forecasting recommendations for sectors.

This document offers solutions to the previously mentioned challenges by suggesting a framework that uses artificial intelligence to improve budget allocation decisions in the Indian public sector. The proposed framework accounts both Union Budget data of the fiscal years 2016-2024 and numerous socio-economic indicators gathered in reliable sources, like the World Bank and the Ministry of Statistics and Programme Implementation. The final decision is a hybrid model using a combination of time-series forecasting and multi-factor regression models to have a greater prediction accuracy.

II. LITERATURE SURVEY

In [1], Jain and Kulkarni (2023) paper examined the Application of AI proposals to the financial forecasting and budgeting strategies. Their analysis in the International Journal of Economics and Management Studies focused on the contribution that predictive analytics, neural networks, and decision support systems can have in improving the accuracy of budgetary planning. The authors showed that AI models would be more responsive to uncertainty and assist in the allocation of resources both in government and corporate finance. They also maintained that conventional forecasting models do not adapt well in dynamic markets.

In [2], Olamijuwon and Zouo (2024) induced an olamijuwon conceptual model of predicting using machine learning, the corporate financial budgeting process. Their manuscript described a ML based Forecasting Framework to assist with budgeting and financial planning through improved accuracy of revenue, expenditure, and investment decisions. While the paper had very little empirical evidence and was more theoretical, it did capture the considerable challenges of AI budget forecasting i.e. data quality, meaningfully interpreting ML models, and ensuring AI based forecasts, integrated into the corporate decision making process.

In [3], Mahmud and colleagues performed a study focused on machine learning-based forecasting in the tourism industry in the United States. They were able to illustrate the manner in which AI models can anticipate market trends, and even consumer spending. The authors used large datasets and regression-based machine learning models, as well as Ensemble Learning, to model consumer behaviour. They concluded that AI-based models can enhance financial planning in sectors that are highly seasonal and repetitive—such as tourism—which can demonstrate very unpredictable buying behaviour.

In [4], AI-Inizi introduced a data-based machine learning-based machine. guided framework of decision support to government. making and policy options. The paper was published in the Babylonian Journal of Machine Learning, it shared they emphasized predictive analytics for polices and budgetary discussions; and illustrated that integrating AI-based forecasting into government options could help to manage economic uncertainty, match spending to fiscal realities, and make the public sector responsive to reality.

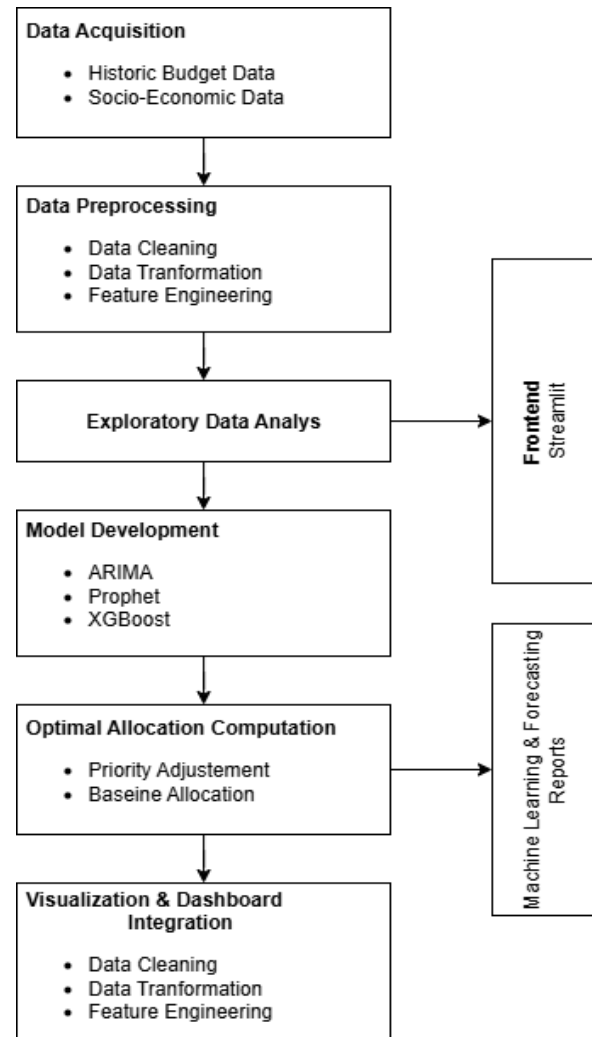
In [5], A study by Talasila (2024)in personal financial management is published in the International Research Journal of Engineering and Technology. The paper highlights the disruptive characteristics of AI for personal financial management through budgeting, tracking spending, and enhancing savings. The author mentions that machine learning algorithms in financial applications may offer individualized recommendations, automatically adapt budget adjustments, and enhance financial literacy among end users thereby interrupting the personal financial planning efforts.

In [6], Adekunle et al. (2021) conducted research for business resource allocation in the Journal of Frontiers in Multidisciplinary Research. The authors used time series models and illustrated how predictive analytics are being implemented across industries in their demand forecasting. The study identified AI-generated value in demand forecasting as derived from the trust factor. The efficiencies

of AI-based demand forecasting are due to waste reduction, increased inventory optimization, and even strategic financial planning. This article provides a framework to complete the connection between demand forecasting and budgeting processes.

In [7], Sun, Feng, and Li (2023) examined an AI-optimized attribution modeling process to facilitate budget allocation in real-time application in parts of the U.S. e-commerce sector. The Journal of Advanced Computing Systems published their work in 2023, and their research resulted in the establishment that budgetary allocation dynamics could be operationalized with the help of real-time analytics. The current research contributed to literature linking data analytics, machine learning, budget allocation, marketing spend, and return on investment, and most interestingly, while the allocator can potentially work in real time, the consumer demand is highly variable in time and space.

In [8], Umronov et al. (2024) presented a forecasting system for economic levels based on machine learning , deep learning , and AI-based systems. The work was published in the 4 th International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE). Their findings show how hybrid AI models can contribute to the prediction of economic levels. The authors noted that the synergy of ML and DL techniques allows for the incorporation of comprehensive financial data in a manageable way to create a strong forecasting capability. Additionally, they found more precision in terms of long-term forecasting which may be economically beneficial in terms of return on investment potential.



III. SYSTEM ANALYSIS AND DESIGN

REQUIREMENT ANALYSIS - As part of the requirement analysis phase of the engagement, the initiative will identify the primary technology and required techno-operational requirements to build the AI-based intelligent budget allocation system for the Indian public sector. The ideal result will be a predictive, adaptive, policy-sensitive fiscal planning instrument using historical data of the Union Budget and socio-economic indicators to generate the best recommendations of the sector-specific budget allocation. The intelligent budget allocation system must provide a reliable projection of the future budget allocation, identify inefficiencies in the existing fiscal planning process, and allow for "what if" scenario simulation.

In order to meet the objectives outlined in the previous discussion, it is assumed that the intelligent budget allocation system will produce output from a structured dataset and the following needs to be prepared. The dataset should be composed of the historical budgets that are organized into sectors from the years 2016 to 2024. The dataset also needs to be able to incorporate macroeconomic and socio-economic indicators which include nominal GDP growth, inflation rate, unemployment, literacy rate, and population statistics.

The data preprocessing process will be required to outline the activities necessary to normalise the currency, remove inflation with constant prices, consolidate expenditures into standardised sector groups, and generate new variables such as ratios of per capita allocations. Each of these steps will have specific requirements that will require a rigorous preprocessing pipeline to ensure data discipline. The predictive modeling component must accommodate hybrid architectures containing both time-series forecasting models (ARIMA, Prophet) for the identification of trends and seasonality and machine-learning regression models (XGBoost and Random Forest) for multi-factor analysis. A hybrid predictive model architecture is one that combines historical trends and social-economic relationships.

ARIMA :

$$\phi(B)(y_t - \mu) = \theta(B)\varepsilon_t \quad (1)$$

Use: Tendency of purchase of budget elements.

Key parts:

$\phi(B)$: Past values (AR).

$\theta(B)$: Past errors (MA).

ε_t : Random noise.

PROPHET :

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \quad (2)$$

Use: Flexible model Trend+seasonality+special events.

Key parts:

$g(t)$: Trend.

$s(t)$: Seasonality.

$h(t)$: Holidays/events.

XGBoost :

$$y_t = f(x_t) + \varepsilon_t \quad (3)$$

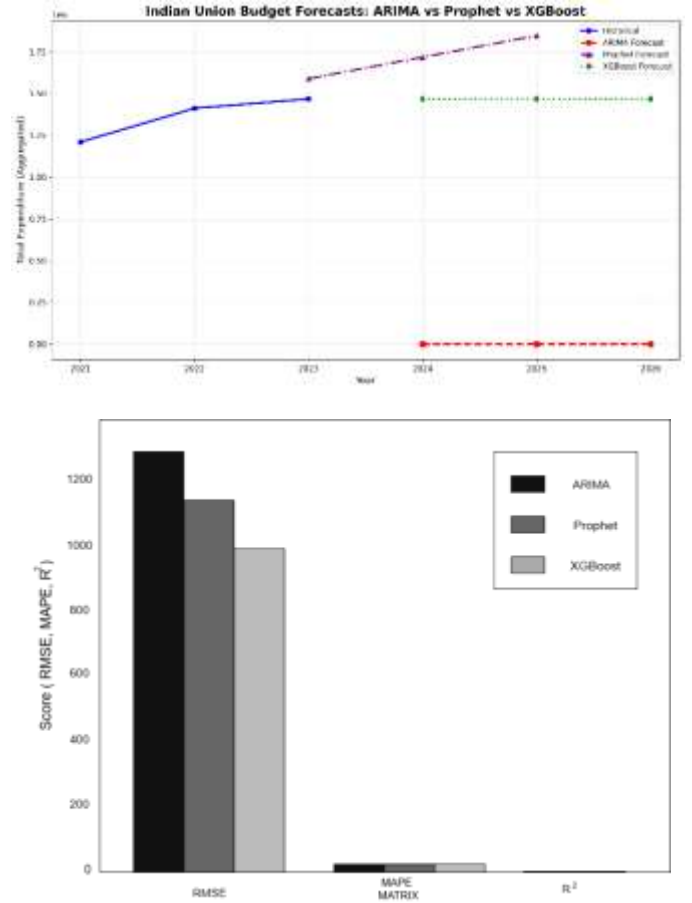
Use: Shows non-linear and multiple relationships between the budget statistics.

Key parts:

x_t : Their characteristics (past values, lagged, externals, such as GDP, inflation, population).

$f(x_t)$ Ensemble of decision trees (boosted).

ε_t : Random error.



We shall use methods of dimensionality reduction and feature selection such as Principal Component Analysis (PCA) to reduce the redundancies and yet capture the economic realities that are of importance. Performance analytics tools will also be used in model assessment to determine the accuracy of predictions, in addition to assessing RMSE, MAPE, R² and other metrics to assess relative performance. Some of the advanced machine-learning applications implemented in the design and implementation of the system will be Python, Pandas, NumPy, Scikitlearn, Statsmodels, and TensorFlow/Keras. The visualization frameworks (e.g. Matplotlib, Plotly or Tableau) will also make it easy to integrate dashboards. Finally, the cloud-based systems (e.g. Google Colab or AWS) will allow contingent power (either in both GPU and CPU) to use large models, as well as allow more immediate data to be used to make decisions during the adjustments of a rolling fiscal year. The system should also be architected for future implementation on the web and in a dashboard format, allowing for policy makers to work with future operating forecasts, change values to the parameters and see changes in the budgets immediately.

ANALYSIS PHASE – Operational Feasibility- The proposed system will be automated to the point that the user will only have to upload historical budget files, and macroeconomic indicators and the system will handle everything else. It will also preprocess, forecast, and recommend/tag KAV for an optimized allocation, with very little human input. This allows central government departments, policy think tanks, and budget travelers to engage in detailed budget analyses even if they lack advanced, discipline-specific technical expertise (i.e., they don't have to have "mathematical modelling skills"). Because it operates

within interactive dashboards, this system could also be integrated into any national-level finance ministry, central state-level budget office, or any remote administration budgets. Future functionality will expand existing consulting brains in cloud-based policy platform, enhancing the usability of coordination and collaboration.

Technical Feasibility- This project is technically feasible as it will be based on gear existing forecasting and regression models. For example, ARIMA, Prophet will catch the temporal trend and the seasonal patterns. The XGBoost and Random Forest models will deal with the more complex socio-economic dependencies that operate in a multi-variable context. Principal Component Analysis (PCA) will assist in trim down the features-space of the potential models in order to not might be losing some signal, while reducing dimensionality. It will be implemented with open-source and completely free software (Python, Pandas, NumPy, Scikit-learn, Statsmodels, and plotting software Plotly/Dash) within a cloud-based system (Google Colab or AWS). This would give space for core computation and model training capacity.

Economic Feasibility- The system will rely on primarily open- source libraries, as well as public datasets that can help reduce the costs of licensing and other expenses of acquiring information. The majority of expenses at the start would come from the effort to collect and sweep the data, along with any computational power needed to train and validate the models. In the end, long-term sustainability will be mainly based on our ongoing costs of maintenance, which we expect to be low, with some predictable expenses related to hosting in the cloud and deploying an advanced dashboard when finished. Given the impact of the system in supporting optimization of allocation patterns to minimize fiscal outstanding inefficiencies, the governments may indeed save a considerable amount of money by avoiding over-funding in any one public sector and ultimately encouraging other critical sectors requiring more financial support.

IV.PROPOSED SOLUTION

Budgeting is one of the most important roles of the national government, to allocate social and government funds and resources to health, education, agriculture, infrastructure, defense, technology, and welfare sectors. In India, this includes not only the allocation of resources but also spending engage the goals of growth and development, reduce inequality, and address serious socioeconomic challenge, and the accountability of ensuring that every rupee is accounted for by determining whether it was the appropriate allocation. Reflecting on a country with more than 1.4 billion people, there are multiple public needs this funding must address, ranging from rural to urban slippage and every level of development state presents unique concern - the diversity is immense. By the time India calculates, bills assessments for or authorizes funding allocation, the limitation in the process does not allow the currently limited consideration of this diversity, as it tends to be common parameters well-established historically have been used as a budget method, which have recurrence and experience to draw on while constraints of options. The budgeting process has accepted common paramenters traditionally and historic proccesses, allows all actors all too easily engage with the practice of the past budgets to base their planing each year, with a base call up procedural practice, which while might hold some method of baseline practice is hardly ever

completely accurate to the current politics of prioritizing schedule in this century.

Moreover, we also wish to refute the notion that public systems, like needed improvement Ontario's, function as intended under evidence and socio-economic principles - for example, we very often behave as if increases in population, under or migration and technology changes or changes in environmental conditions don't created the need to reassess and readjust those public systems and the budgets that drive public expenditure. When a public system relies heavily on historic public use data will create change inertia in funding patterns. That is to say, funding may flow to public facilities or projects that have historically experience funding in spite of urgent need and funding may administer to the older projects or sectors in need while some vast urgent needs are foregone. For example, if a state experiences a moderate degree of industrialization, there will be an urgent need for infrastructure funding, while funding still flows to states that have historically contracted with more funding.

Another limitation is weak forecasting systems. There are numerous reactive periods, which react to the existing obstacles, rather than preventive investment. Without forecasting metrics, it is possible for the government to misread needs in priority areas. These reactivity pressures become problematic in crises created by pandemics, or natural disasters, or economic downturns. When unforeseen events take place, funds must be diverted to urgent needs and the programmatic focus of blunt seated policies is disrupted between sectors. This results in time losses and incongruity with levels of government.

This concern is worsened by the lack of access to allocation data. While some amounts can be accessed by citizens, the data is usually in a format that's incredibly difficult for anyone to understand. This, in turn, discourages citizens from participating in budgeting, which leads to even less oversight. There are also issues of lack of a co-ordinated data system which pose a challenge in terms of transparency. Economic, demographic, environmental, and social data are often stored in various ministries and agencies, in compartmentalized spaces with little interconnectivity. As a result, it becomes difficult to have a coherent and contemporary perspective of the needs of the nation. Policymakers often make decisions on the basis of past experiences and information, that may not fully represent the full situation. This is manifestation of incorrect priorities allocation, which, in most cases, leads to money flow into the areas of less urgency, whilst the areas that require funding the most are left without it. Manual-occurring processes can also impede or slow budgetary planning; although the human touch is essential, relying on human discretion provides an opportunity to inject a delay/error through automatic calculations, paper flows, and traditional mumbo jumbo. Additionally, these efficiencies and inefficiencies costs cash, particularly in the budgetary space, given that socio-economic factors are not static. Certainly manual-occurring processes obstruct running meaningful series of simulations on potential budgetary options. In today's governance environment, it is imperative to be able to model potential impacts of different allocations; however, it is virtually impossible to go beyond the existing status quo in India due to technical shortcomings and hope to achieve this level of sophisticated planning capability.

It is always difficult to make sure that the national mandates or laws do not affect the local priorities unnecessarily. States and districts have varying development priorities, and these are often in conflict with national efforts or initiatives. If we are not given "flexible" allocation models or data models to make changes, it

can be a difficult situation. Current economic disparity makes balancing these types of relationships a greater challenge. Wealthy states, and their districts, can get a greater piece of the pie, while the poorer states continue to get left behind. They result in sluggish and inclusive development and increasing regional disparity.

Political influences also affect budget distributions and they obviously favor locations close to election time. Referring to a nexus of activity, cooperation, or political and operational activity with respect to deliberation, management, and an outcomes; as it goes too far, it would produce an inappropriate bias that would impair fairness and ability, local fairness and ability, etc. A consistent absence of allocation using AI systems prevents budget/course of action specialists access to the best analytic tool possible to analyze commodities and to glean forecasts on future commodities or uses. Today's AI could integrate large datasets, identify unusual trends, and prescribe the optimal distribution models for everything from the short-term effective ways and change in lots as well.

Urbanization and population growth increase the demands placed on public infrastructure, health care, housing and other public needs. Unless we act to improve budgeting, and we have no time to make good data and evidence-based choices, we are going to face the greater over-concentration and unfair downgrading of the services to the population, and widen the economic imbalance between urban and rural communities. The issue of climate change is also exerting pressure on a complicated situation. Climate change and natural disasters that come with the alteration of weather conditions (floods, droughts, high temperatures, etc.) demand a budget that is flexible in handling the extreme conditions and the dynamism of such conditions. The inflexible system which we now possess cannot permit fluidity which is at times demanded by changing conditions.

The budget allocation processes in India are a very complex combination of a set of challenges: the outdated approaches; the inability to gather the wholesome and consistent data; the absence of transparency, opaqueness, and unfair justice toward power; the deficiencies in technologies; the lack of the predictive planning. Existing models cannot overcome these challenges, but they need a transition to AI-driven, data-integrated, and optimization-driven processes, which will be oriented to equitable transparent and futureproof governance and budgetary allocation.

V. PROJECT DESCRIPTION

To overcome the limitations of the traditional allocation protocols that are largely dependent on previous spending patterns and marginal variations, the proposed project will help to develop and implement an AI-based application to improve budget allocation process in the Indian public sector. The approach outlines the predictive, adaptive and policy based budget planning approach by overlaying the past Union Budget data which spans 2016-2024 superimposed with the macroeconomic variables (GDP growth, inflation rates, open unemployment rate, literacy rate, and population size) in each of the sectors. Its processing method begins with a giant preprocessing data operation to form new features, e.g. per capita allocations, lagged financial variables to accommodate temporal dependencies, standardize financial variables, group expenditures into sector-driven features, and use constant

prices to account for inflation.

In a hybrid design, the predictive modeling component is comprised of statistical methods that support and utilize machine learning. The univariate time series forecasting models, such as ARIMA, Prophet, to predict pattern and seasonality, and effective machine learning regression models, such as Random Forest and XGBoost, can predict socioeconomic indicators using budget allocations and the time series, at the sector level, bounded by time series forecasting methods. Districts can capitalize on time series analysis within a multivariate socioeconomic model simultaneously.

By blending the strengths of univariate time series analysis, where it can be flexed to its fullest, with the augmenting interpretative power of the multivariate socioeconomic information, another plane of clarity and precision of prediction modeling is created. Redundancy measure can be evaded and can result in better computational time and efficiency, provided that the accuracy of the predictions is not compromised by the feature selection of and utilization of the features as dimensions via dimensional reduction techniques such as thrown dimensionality reduction with PCA.

The framework includes scenario models that give the practitioners a chance to play with the what- if scenarios of the allocation deliberations (i.e. what happens in the event of the budget parameters or policy intent or economic assumptions) and then instantly spend the result of the thought on the allocation. After developing a total expenditure forecast the system uses a weight-adjustment optimization model and a level of priority ranking to assist in the reallocation of resources within the budget constraints and policy purpose. The simulation outcomes are published through interactive visualization dashboard which facilitates accountability and transparency because policy-oriented individuals access the evidence, the rationale and purpose behind each budget submission.

The project develops a collaborative decision-support system, which can be scaled and applicable across a large range of governance levels (national and state ministries, municipal governments) that combines econometric forecasting, machine learning-based regression and optimization-based models of allocation in a single system. It is meant to be cheap and thus is based on the use of open source and publicly available data. All in all, the advantages of this decision support tool are a decrease in unnecessary spending of the public, more directly planned distributions in accordance with the socio-economic priorities of the population, a greater level of transparency in fiscal plans, and the possibility of the policymakers to implement the evidence-based decisions in the rapidly evolving social and economic environment. Thus, this framework is in the hope that it will facilitate a more responsive, efficient, and equitable public system of financial management as a result of India being able to respond to being more inclusive and sustainable in socioeconomic growth.

	RMSE	MAPE	R ²
ARIMA	1200	8.5%	0.82
Prophet	1100	7.9%	0.85
XGBoost	950	6.3%	0.91

VI. CONCLUSION

The AI-based budget allocation prediction system outlined here addresses a major weakness in India's financial planning approach, which is to replace traditional, static budget allocation methods with an ever-changing, data-driven approach. This system will enable budget allocations to be properly budgeted and re-allocated to the shift in social priorities and requirements by incorporating machine learning models, socio-economic and other indicators and past historical data on financial allocations. Such a paradigm shift can significantly enhance the effectiveness of the decision-making process, the effective utilization of the resources as well as the effectiveness of reaction to government spending.

The inherent scalability and modular nature of the system in all levels of government, both in the central and a local administrative body is a strength of the system since it will enhance the predictability ability of policy makers to future needs. This kind of accurate prediction will mean that only where there is maximum socio-economic benefit will funding be made. Moreover, since the system will be equipped with the ability to simulate how approaches to distribution of funds can be developed, the system will enable decision makers to discover trade-offs and develop more confidently established funding/prioritize initiatives based on, more than just assumption based reasoning as a method of estimating priorities.

The system is reflective of transparency. It implements new analytic and visual dashboards whereby all stakeholders can view and. Real time monitoring allocation decisions and justification. This is an accountability factor. With new technologies like blockchain, it might be in the future a manifestation of. Record of events of the allocation, which is immutable, and that can concurrently give future popular trust in its. Financial accountability. Consequently, the goal is to restrict. The possibility of abuse, and give an account of allocations. To prove that resources have been dedicated either openly or equitably or inequitably.

The social change attribution of this solution would have a broader influence than the enhancement of governance of the process of allocating it. It is hoped that it will be used to make proactive responses, on the basis of needs, to aid in decreasing the incremental gaps in the health care, educational, infrastructure, and social welfare. With the solution, the underprivileged communities have to be put in a better position, and help the growth of the region through a wider formed direction, which is interconnected with strategic national development objectives. The solution must serve as a converging system, which other nations will then use artificial intelligence to modernise their budget systems at some point in time.

In conclusion, the event of launching the AI-based budget allocation forecasting system and its implementation is a definite turning point in the current progress of the field of the public financial management. Not only does it promote efficiencies and effect the budget allocation process but it also aims at showing respect to principles of fairness, transparency and long-term sustainability. As an illustration, it reconciles the old paradigm of policymaking in the governmental sector with the new citizen-based data-driven decision-making, which holds promising opportunities to convert the budget planning approach to responsive, proactive, citizen-based planning to the future-oriented

sustainable and outcomes-oriented growth of the Indian objective of inclusive growth. Moreover, the modular, scalable quality that translates to the fact that it can be used at any level of government in the form of policies to locality in terms of local policy-making. There is also progress being made towards an advanced concept, incorporating IoT and Blockchain, that will provide even more levels of transparency and accountability, in case/when an examination of historical data when used in conjunction with real time data based on new technologies become more widely available in publicly accountable systems, society may look forward to having the ability to use the transformational potential of AI in the government budgeting and spending to establish trust and knowledge in the moral and equity of government budgets and government spending.

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